

CSTR with Elementary Reversible Liquid-Phase Reaction

Problem P12–16b: Model Development and Analysis

1 Problem Description

We consider the elementary reversible liquid-phase reaction



taking place in a continuous stirred-tank reactor (CSTR) equipped with a cooling jacket (or heat exchanger). Pure A enters the reactor at flow rate F_{A0} and temperature T_0 ; the coolant (or ambient) is at temperature T_a . The reactor volume is V , and the volumetric flow rate is v_0 so the space time is

$$\tau = \frac{V}{v_0}. \quad (1)$$

The overall heat-transfer coefficient–area product is UA .

The reaction is elementary and reversible. At 400 K the kinetic and equilibrium parameters are

$$k(400 \text{ K}) = 1 \text{ min}^{-1},$$
$$K_C(400 \text{ K}) = 100,$$

and the activation parameter and heat of reaction are

$$\frac{E}{R} = 2.0 \times 10^4 \text{ K},$$
$$\Delta H_R = -8.0 \times 10^4 \text{ cal mol}^{-1} \quad (A \rightarrow B, \text{ exothermic}).$$

The heat capacity of each species is taken as constant: $C_{P,A} = C_{P,B} = 40 \text{ cal mol}^{-1}\text{K}^{-1}$.

We develop expressions for the heat-generation–heat-removal function $G(T)$, determine steady states and their stability, and explore the effect of ambient temperature, heat-transfer coefficient and flow rate.

2 Kinetics and Equilibrium

2.1 Temperature dependence of the rate constant

The forward rate constant $k(T)$ follows an Arrhenius form referenced to 400 K:

$$k(T) = k_{400} \exp \left[-\frac{E}{R} \left(\frac{1}{T} - \frac{1}{400} \right) \right], \quad (2)$$

where $k_{400} = 1 \text{ min}^{-1}$ and $E/R = 2.0 \times 10^4 \text{ K}$.

2.2 Temperature dependence of the equilibrium constant

The equilibrium constant is given at 400 K as $K_{C,400} = 100$. Assuming the van't Hoff relation for a constant heat of reaction ΔH_R ,

$$\frac{d \ln K_C}{dT} = \frac{\Delta H_R}{RT^2}, \quad (3)$$

we integrate between T and 400 K to obtain

$$K_C(T) = K_{C,400} \exp \left[-\frac{\Delta H_R}{R} \left(\frac{1}{T} - \frac{1}{400} \right) \right]. \quad (4)$$

Because $\Delta H_R < 0$ (exothermic), K_C increases with temperature.

2.3 Rate expression

With an elementary reversible reaction in a well-mixed liquid phase, the net rate of consumption of A is

$$-r_A = k(T) \left(C_A - \frac{C_B}{K_C(T)} \right). \quad (5)$$

3 Stoichiometry and Concentrations

The feed contains only A with concentration $C_{A0} = F_{A0}/v_0$. Let X be the conversion of A in the reactor:

$$X = \frac{F_{A0} - F_A}{F_{A0}}. \quad (6)$$

For the isothermal liquid-phase system with $A \rightleftharpoons B$ and no change in total moles, the outlet concentrations are

$$C_A = C_{A0}(1 - X), \quad (7)$$

$$C_B = C_{A0}X. \quad (8)$$

Substituting into Eq. (5) gives

$$-r_A = k(T) C_{A0} \left[1 - X \left(1 + \frac{1}{K_C(T)} \right) \right]. \quad (9)$$

4 CSTR Steady-State Mole Balance

For A in a CSTR at steady state,

$$F_{A0} - F_A + V r_A = 0 \quad \Rightarrow \quad F_{A0}X = V(-r_A). \quad (10)$$

Using Eq. (9) and recognizing that $F_{A0} = C_{A0}v_0$ and $\tau = V/v_0$,

$$F_{A0}X = V k(T) C_{A0} \left[1 - X \left(1 + \frac{1}{K_C(T)} \right) \right], \quad (11)$$

$$C_{A0}v_0X = C_{A0}v_0\tau k(T) \left[1 - X \left(1 + \frac{1}{K_C(T)} \right) \right]. \quad (12)$$

Cancelling common factors,

$$X = \tau k(T) \left[1 - X \left(1 + \frac{1}{K_C(T)} \right) \right]. \quad (13)$$

Collecting terms in X ,

$$X + \tau k(T) \left(1 + \frac{1}{K_C(T)} \right) X = \tau k(T). \quad (14)$$

Hence the conversion as a function of temperature is

$$X(T) = \frac{\tau k(T)}{1 + \tau k(T) \left(1 + \frac{1}{K_C(T)} \right)}. \quad (15)$$

This relation couples reaction kinetics and equilibrium with the reactor residence time.

5 Energy Balance and the Heat-Generation Function

5.1 Steady-state energy balance

Assuming constant density and heat capacity, no shaft work, and a single outlet stream, the steady-state energy balance for the CSTR is

$$0 = F_{A0}C_P(T_0 - T) + (-\Delta H_R)F_{A0}X(T) - UA(T - T_a). \quad (16)$$

The first term is the sensible enthalpy change between the inlet and outlet streams, the second term is the heat released by reaction, and the third term is the heat removed by the heat exchanger.

5.2 Definition of $G(T)$

It is convenient to rearrange the energy balance into the form

$$G(T) \equiv F_{A0}C_P(T_0 - T) + (-\Delta H_R)F_{A0}X(T) - UA(T - T_a) = 0. \quad (17)$$

The function $G(T)$ represents the difference between the rate of heat generation and the rate of heat removal:

$$G(T) = Q_{\text{gen}}(T) - Q_{\text{rem}}(T). \quad (18)$$

Steady-state reactor temperatures are the roots of $G(T)$.

Substituting Eq. (15) together with Eqs. (2) and (4) into Eq. (17) provides an explicit expression for $G(T)$ in terms of known parameters.

6 Steady States and Local Stability

6.1 Steady states

For a given set of parameters $(UA, T_a, v_0, \dots)$, the steady-state temperatures are the real solutions of

$$G(T) = 0. \quad (19)$$

These are obtained numerically in MATLAB using the built-in root finder `fzero`, starting from several initial guesses to locate up to three distinct roots. For the nominal case ($UA = 3600 \text{ cal/min/K}$, $T_a = 310 \text{ K}$, $v_0 = 1 \text{ dm}^3/\text{min}$) three steady states are found near

$$T_1 \approx 310 \text{ K}, \quad T_2 \approx 377 \text{ K}, \quad T_3 \approx 418 \text{ K}. \quad (20)$$

The corresponding conversions follow from Eq. (15).

6.2 Local stability criterion

The unsteady-state energy balance can be written schematically as

$$\rho C_P V \frac{dT}{dt} = G(T), \quad (21)$$

where $\rho C_P V$ is the lumped thermal capacitance of the reactor contents. At a steady state $T = T^*$ the time derivative is zero and $G(T^*) = 0$.

Linearizing about T^* ,

$$\frac{d}{dt}(T - T^*) \approx \frac{1}{\rho C_P V} G'(T^*) (T - T^*), \quad (22)$$

where $G'(T^*) = \left. \frac{dG}{dT} \right|_{T^*}$. The eigenvalue of the linearized system is

$$\lambda = \frac{G'(T^*)}{\rho C_P V}. \quad (23)$$

Hence:

- If $G'(T^*) < 0$ then $\lambda < 0$ and perturbations decay in time: the steady state is *locally stable*.
- If $G'(T^*) > 0$ then $\lambda > 0$ and perturbations grow: the steady state is *locally unstable*.

Numerically, $G'(T^*)$ is approximated by a centered finite difference, and the sign is used to classify each steady state.

For the nominal case, the lowest and highest temperature steady states are stable, while the intermediate one is unstable, yielding the characteristic “S-shaped” multiplicity diagram for an exothermic CSTR with cooling.

7 Parametric Studies

7.1 Effect of ambient temperature T_a (ignition and extinction)

To study ignition and extinction, we treat T_a as a parameter and, for each chosen value, compute the steady-state temperatures by solving $G(T) = 0$. Plotting the resulting steady states T versus T_a gives an S-shaped curve. The turning points of this curve correspond to:

- *Ignition temperature*: the value of T_a at which the low- and middle-temperature steady states coalesce and disappear.
- *Extinction temperature*: the point where the high- and middle-temperature branches merge.

These points mark abrupt transitions between low-conversion and runaway branches during startup and shutdown.

7.2 Heat-exchanger failure ($UA = 0$)

If the heat exchanger fails, we set $UA = 0$ in Eq. (17). The energy balance becomes

$$F_{A0}C_P(T_0 - T) + (-\Delta H_R)F_{A0}X(T) = 0. \quad (24)$$

Using Eq. (15), the new steady states are obtained by solving this equation numerically. For the parameters given, the upper steady-state temperature increases to approximately 431 K, with a corresponding conversion on the order of a few percent.

7.3 Effect of UA and maximum conversion

Treating the product UA as a design variable, the procedure is repeated for different UA values. For each UA , all steady states are located and the maximum achievable conversion among them is recorded. This yields a curve $X_{\max}(UA)$. There is an optimal UA where this maximum conversion is largest: if UA is too small, inadequate cooling causes runaway; if UA is too large, the reactor operates close to the feed temperature with little conversion. The MATLAB script estimates this optimal UA to be in the range $7\text{--}8 \times 10^3 \text{ cal min}^{-1}\text{K}^{-1}$ with a maximum conversion of roughly 85%.

7.4 Adiabatic blowout flow rate

For adiabatic operation ($UA = 0$), the energy balance simplifies to

$$C_P(T_0 - T) + (-\Delta H_R)X(T; \tau) = 0, \quad (25)$$

where $X(T; \tau)$ is given by Eq. (15) with $\tau = V/v_0$. Because X depends on τ , the steady-state temperature depends on the flow rate v_0 . For certain parameter combinations the reduced balance (25) can admit multiple steady states in T ; the *adiabatic blowout* condition corresponds to a critical residence time τ_{crit} where two steady states coalesce in a saddle-node bifurcation, i.e.

$$G_{\text{ad}}(T; \tau) = C_P(T_0 - T) + (-\Delta H_R)X(T; \tau) = 0, \quad \frac{\partial G_{\text{ad}}}{\partial T} = 0. \quad (26)$$

In practice, the critical τ can be approximated numerically by scanning over τ values, counting the number of steady-state roots and identifying where this number changes. The corresponding flow rate is

$$v_{0, \text{blowout}} = \frac{V}{\tau_{\text{crit}}}. \quad (27)$$

8 Safe Operation at the Lower Steady State

To operate safely at the lower-temperature steady state and avoid thermal runaway to the upper steady state, the design and operating conditions should be chosen so that the system remains far from the ignition point. Strategies include:

- Increasing the heat-removal capacity UA so that only a single, low-temperature steady state exists.
- Lowering the feed temperature T_0 and/or ambient temperature T_a .
- Reducing the feed concentration C_{A0} or molar feed rate F_{A0} to limit heat generation.